

A Mini Project Report entitled on
SMART SHOPPING TROLLEY WITH AUTOMATED BILLING

Submitted in partial fulfillment of the requirements for the award of the
degree of

Master of Engineering

in

Embedded Systems

By

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

Stanley College of Engineering and Technology for Women (Autonomous)

**(Approved by AICTE, Accredited by NBA and NAAC, Affiliated to
Osmania University)**

Chapel Road, Abids, Hyderabad

2024-25



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FOR WOMEN (AUTONOMOUS)**

Chapel Road, Abids, Hyderabad – 500001

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PS02: Adaptability to Productive Environment: To be well equipped with Management skills, interdisciplinary and modern technologies.

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CERTIFICATE

This is to certify that the MINI PROJECT title entitled “**SMART SHOPPING TROLLEY WITH AUTOMATED BILLING**” is carried out by **KHATIJA MAHVEEN (160624744407)** in partial fulfillment for the award of degree of **MASTER OF ENGINEERING** in Electronics and Communication Engineering, Stanley College of Engineering and Technology for Women (Autonomous), Abids, Hyderabad, during the academic year 2024-25.

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DECLARATION

I declare that the work reported in the thesis entitled “**SMART SHOPPING TROLLEY WITH AUTOMATED BILLING**” is a record of the work done by me in the Department of Electronics and Communication Engineering, Stanley College of Engineering and Technology for Women, Hyderabad.

Khatija Mahveen

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ABSTRACT

With the growing demand for automation in everyday life, especially in retail environments, there is an increasing need for smart systems that can improve operational efficiency and customer convenience. One of the major bottlenecks in supermarkets is the traditional billing system, which often results in long queues, delayed checkouts, and manual errors. This project proposes the development of a Smart Shopping Trolley with Automated Billing, aimed at transforming the conventional shopping experience into a more seamless, time-efficient, and user-friendly process.

The system is embedded with electronic components that enable automatic detection and billing of products during shopping itself, reducing the need for customers to wait at billing counters. It reflects an integration of embedded technology with retail practices, aligning with the vision of smart retail infrastructure. This innovation not only enhances billing accuracy and reduces manpower dependency but also offers a scalable solution for busy retail environments.

The project stands as a practical model that addresses real-world problems through a combination of affordability, automation, and user interaction, paving the way for intelligent retail systems of the future.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In the age of rapid technological advancement, automation has become a key driver of innovation across various domains. From homes to industries, smart systems are reshaping the way humans interact with their surroundings. One area that has seen significant technological integration in recent years is the retail sector. Shopping malls and supermarkets are vital parts of modern consumer life, serving as essential hubs for daily needs. However, the traditional functioning of these environments still relies heavily on manual processes, especially during checkout and billing. This often leads to long queues, increased waiting times, human errors, and customer dissatisfaction.



Fig 1.1- Groceries Checkout counter

To address such challenges, the concept of smart retail and intelligent shopping systems has emerged. These solutions aim to automate and streamline common customer tasks such as product selection, billing, and checkout. Among these, smart shopping trolleys stand out as an innovative tool to directly assist customers while shopping. By integrating basic electronics and embedded systems into a traditional trolley, such devices can enable real-time product scanning, dynamic billing updates, and efficient item management all from within the trolley itself. This drastically reduces the customer's dependency on billing counters and cashier systems, making the shopping experience faster and more pleasant.

This project, titled “Smart Shopping Trolley with Automated Billing”, has been developed as a practical application of embedded systems to improve the retail shopping experience. It provides a solution that allows users to scan items as they shop and receive live billing feedback through an interactive LCD screen.

This integration of hardware and software makes it possible for customers to keep track of their purchases instantly without relying on external human interaction or infrastructure. Moreover, by using RFID technology, the system ensures accuracy and efficiency in identifying products, further eliminating the possibility of billing mismatches; the same concept can be extended in the future with wireless modules to connect to cloud databases, inventory systems, or mobile applications and consumer convenience into one cohesive solution. It addresses key inefficiencies in current shopping systems and serves as a prototype for the future of smart retail.

The choice of components such as the Arduino Nano, RFID reader module (RC522), 16x2 LCD with I2C, and rechargeable battery systems has been made with portability and cost-effectiveness in mind. These components not only reduce the overall complexity of the system but also make it feasible for real-world implementation in small-scale and large-scale retail outlets. The Smart Shopping Trolley project is also a step toward integrating IoT (Internet of Things) concepts in retail, laying the foundation for more advanced smart retail ecosystems. While this version of the trolley operates offline and locally, the same concept can be extended in the future with wireless modules to connect to cloud databases, inventory systems, or mobile applications. The evolution of such systems can pave the way for complete cashier-less stores, which are being experimented with by global tech companies.

The Trolley is a forward-thinking application that merges automation, electronics, and consumer convenience into one cohesive solution. It addresses key inefficiencies in current shopping systems and serves as a prototype for the future of smart retail. Its simplicity in design, combined with high impact in functionality, makes it a viable option for real-world implementation. With further development, this system can evolve into a fully integrated smart retail model aligned with the vision of digital transformation.

1.2 Problem Specification

1. Traditional supermarket billing systems require customers to wait in long queues, leading to time consumption and customer dissatisfaction.
2. Manual billing is prone to human errors in price entry, item count, and product recognition.
3. Lack of real-time billing feedback prevents customers from tracking their expenditure while shopping.

1.3 Objectives

1. To design and develop a smart shopping trolley that automates the billing process using RFID technology.
2. To enhance customer convenience through self-checkout and an interactive LCD display for real-time item and billing updates.
3. To improve billing accuracy and reduce human errors through a fully embedded, user-friendly system.
4. To minimize customer waiting time at billing counters by enabling real-time, on-the-go billing during shopping.
5. To demonstrate low-cost, scalable solutions suitable for implementation in modern supermarkets

1.4 Advantages

1. Reduces billing time
2. Improves accuracy
3. Enhances customer experience
4. Reduces manpower dependency
5. Portable and user-friendly design
6. Scalable for future expansion
7. Low cost
8. Provides flexibility
9. Easy to implement

1.5 Methodology

The methodology adopted for the development of the Smart Shopping Trolley with Automated Billing involves a systematic process that integrates both hardware and software aspects of embedded system design. The process begins with understanding the key objectives and functional requirements such as automated item detection, live billing updates, and ease of user interaction.

Based on these requirements, suitable hardware components including the Arduino Nano microcontroller, RC522 RFID reader module, 16x2 LCD display with I2C interface, a green push button, buzzer, rechargeable battery, and a TP4056 charging module are carefully selected for cost-efficiency and reliability. A detailed block diagram is designed to visualize the overall architecture and data flow between components. The components are then assembled and connected on a dot PCB board, with proper pin configurations, ensuring compactness and portability.

The software development is carried out using Arduino IDE, where custom logic is written to handle multiple operations like RFID tag scanning, storing item IDs, updating total cost dynamically, detecting button presses for item removal, and displaying information on the LCD. Once the code is compiled and uploaded to the microcontroller, the system is tested with various RFID tags to simulate real-world shopping scenarios. Special attention is given to the functionality of removing an item using the push button during re-scanning. A detailed block diagram is designed to visualize the overall architecture and data flow between components. The components are then assembled and connected on a dot PCB board, with proper pin configurations, ensuring compactness and portability.

Debugging and iteration are done at both hardware and software levels to ensure stable performance. Finally, the complete working prototype is mounted onto a physical trolley frame, representing a real shopping environment, where the system is evaluated for its efficiency, speed, and user convenience. The outcome of this methodology is a functional and user-friendly prototype that reflects the potential of embedded automation in retail shopping applications is given to the functionality of removing an item using the push button during re-scanning. Debugging and iteration are done at both hardware and software levels to ensure stable performance.

1.6 Layout of the Report

The report explains the implementation of “Smart Shopping Trolley With Automated Billing”.

Chapter 1: Presents introduction to overall report and overview of the project.

Chapter 2: Literature Survey on different existing methods and systems.

Chapter 3: It deals with the block diagram of the project and explains the purpose of each block.

Chapter 4: It deals with Hardware description.

Chapter 5: It explains the software implementation of the project using Arduino IDE.

Chapter 6: This chapter contains results of the proposed system.

Chapter 7: This chapter contains Conclusion and Future scope of project.

CHAPTER 2

LITERATURE SURVEY

[1]. Rajkumar Patil, Alim Bahadur, Abhinandan Kandekar, Shubham Koli, Shridhar Bosge. "Automatic billing system by using artificial intelligence." International Research Journal of Modernization in Engineering Technology and Science, 2021.

In this paper, a fully automated billing system is proposed which is built by integrating mechanical conveyor belts with artificial intelligence (AI), which will overcome the limitations of the traditional system. In this concept, customers need to put their things one by one on the conveyor belt. The conveyor belt includes a high resolution camera to take photos of objects passing by, and then recognize the objects with the help of object detection algorithms and automatically generate Bills. This AI based algorithm uses a custom data set of the product to be billed. The data set contains the number of images of the product and the price of the product. The main advantage of this technology is that the orientation of the object does not matter in detection. The model can detect the object in any of its orientations.

[2]. Mrudul Padole, Apurva Gupta, Ayush Kumar, Vitthal Gadekar, Prof. S.R. Patil. "Smart Supermarket Billing System Using Python." IJASRET, 2021.

The provided excerpt discusses the intention to use the Java programming language and MySQL database to develop a framework for automating the billing process in supermarkets, aiming to contribute to India's digital transformation database, represents a significant step towards enhancing efficiency and customer experience in the retail sector. Here's a conclusion based on the information provided The proposed framework for automating the billing process in supermarkets, developed using Java programming language and MySQL database, represents a significant step towards enhancing efficiency and customer experience in the retail sector. By leveraging technology to streamline the billing process, the framework aligns with the objectives of India's digital transformation initiatives. Customers are likely to benefit from a smoother shopping experience, while management can expect improved operational efficiency and data management capabilities.

[3]. Namitha James, Nikhitha Theresa Antony, Sara Philo Shaji, Sherin Baby, Jotsna Annakutty. "Automated Checkout for Stores: A Computer Vision Approach." Student, Department of Computer Science and Engineering, Rajagiri School of Engineering and Technology, Kerala, India, 2021.

The proposed method detects the objects in the MS COCO dataset which includes items from the hypermarket. Then it counts the number of similar items, generates the bill and payment details in a GUI interface. In the proposed model's training procedure, the data preparation step and the model training phases were separated. During the data preparation phase, data labeling and preprocessing were done to prepare the training data.

[4]. Martinus, Metta Saridewi Wahab, Yudi, Hanry Ham. "Data Transmission Using RFID System on Smart Shopping Carts for Checkout Process Efficiency in Supermarkets." ScienceDirect, 2021.

Products in the supermarket are equipped with RFID tags containing item IDs and prices. Customers tap these RFID-tagged items to a reader in their smart shopping cart, which sends the data to the server. The server updates a database with transaction information and synchronizes it with the customer's mobile application. When customers are ready to complete their transaction, they show a one-time password displayed on their app to the cashier. The cashier inputs the code into the web application to retrieve the transaction details for final check and payment processing. Once the transaction is complete, the cashier clicks the finish button on the web application to inform the server, which updates the database accordingly.

[5]. Suraj Chopade, Smita Palnitkar, Sujit Chavan, Anirudha Deshpande. "Automated Super Shop using Image Processing (Python)." IJFGCN International Journal of Future Generation Communication and Networking, 2020.

Describes supermarket billing through automated processes. Utilizing a database of predefined shapes, a camera captures images of goods and detects these shapes, comparing them with stored data. The software then calculates the bill amount based on recognized objects and their corresponding prices. For registered customers, the bill can be automatically debited from their accounts, enhancing convenience. Additionally, the system incorporates a mobile application offering users features like

product Searching, preordering, and seamless online payments, providing a comprehensive and efficient shopping experience.

[6]. P. Kalaivani, C. Rajan, S. Sandhiya. "An Automated Billing System in Shopping Malls Using Bascart." Bulletin of Scientific Research, 2019.

To streamline the billing process, a Smart Trolley System (STM) utilizing Wireless Access Network and RFID technologies is proposed. Connected to the store's router, customers log in to the store's website to access their list of purchased items. The Smart Trolley, equipped with an RFID scanner, automatically scans items with RFID tags upon placement, updating the total quantity and amount displayed on an attached LCD screen. Customers can conveniently remove purchased items, with the total adjusting accordingly. Once shopping is complete, customers proceed to pay the displayed total amount and logout from the store's website.

[7]. Priyanka S. Sahare, Anup Gade, Jayant Rohankar. "A Review on Automated Billing for Smart Shopping System Using IOT." IJETA, 2019.

The objective of the future system is to provide a low-cost and easily accessible shopping support system. The RFID-powered electronic shopping trolley enhances the shopping experience for computer electronics store consumers. Customers scan items using the trolley's RFID reader, which records the item's cost. Upon punching their RFID card, the Raspberry Pi retrieves the item details from the database and displays them on an LCD, along with the total cart amount, aiding customers in budget management. Pressing a key on the trolley automatically sends the data to the billing counter, expediting the checkout process. This system streamlines shopping, ensures transparency in pricing, and improves overall customer satisfaction.

[8]. Mrs. Swetha, Ms. Veena Shellikeri, Mr. Muneshwara, Dr. Thungamani. "Survey of Object Detection Using Deep Neural Networks." 2018.

The training methodology involved utilizing a dataset comprising 10 million crops overlapping with objects with at least 0.5 Jaccard overlap similarity, each labeled with one of the 20 VOC object classes. An additional 20 million negative crops were included, labeled with the "background" class label. For evaluation, the localizer model was applied to the maximum center square crop in the image, resized to a 220 x 220 input size. Following non-maximum suppression with a 0.5 overlap threshold,

the Highest scoring detections were retained. These detections underwent classification by the 21-way classifier model, and the final detection score was computed as the product of the localizer for the given box and the classifier score evaluated on the maximum square region around the crop.

[9]. D. S. V. Siva Vardhan, Y. Shivraj Narayan. "Development of an Automatic Monitoring and Control System for the Objects on the Conveyor Belt." 2015.

Belt conveyors are durable and reliable components used in the industries for transporting material. Conveyors producing a single type of component at a time generally monitor and control a single type of object. This paper presents the design, fabrication and validation of an automatic monitoring and control system designed for probing various types of objects on a belt conveyor. This system uses Allen Bradley MicroLogix PLC for controlling and feedback of sensors for monitoring the objects.

CHAPTER 3

PROPOSED SYSTEM

3.1 Existing System

The traditional supermarket billing system heavily relies on manual processes where customers are required to stand in long queues at billing counters after completing their shopping. These systems often involve barcode scanners and human cashiers, which increases the chances of billing delays, manual errors, and customer dissatisfaction.

The proposed Smart Shopping Trolley system introduces automation directly at the customer's point of interaction. By integrating RFID technology, real-time billing display, and embedded electronics within the trolley itself, this system minimizes human intervention, speeds up the shopping experience, and enhances billing accuracy. Unlike traditional systems, the proposed model empowers users to manage their own cart, track their total bill instantly, and even remove items without waiting for staff assistance. The table below summarizes the key differences between the existing system and the proposed smart trolley system.

Hence to overcome these drawbacks we have designed “The Smart Shopping Trolley with Automated Billing” that integrates RFID (Radio Frequency Identification), sensors, and a screen display to enable real-time billing, reduce human error, and offer a faster, smarter, and more convenient shopping experience.

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Feature	Existing System	Proposed System
Billing Method	Manual (Cashier operated)	Automated(Self-service with RFID)
Queue Requirement	Long queues at billing counters	No queue, billing done while shopping
Item Detection	Barcode scanning by staff	RFID tag detection by microcontroller
User Interaction	Passive (staff handles everything)	Active
Real-Time Bill Display	Only at final counter	Continuously updated on LCD screen
Item Removal Flexibility	Limited, requires staff help	Instant by pressing button and re-scanning
Dependency on Manpower	High	Very Low
Error Possibility	High (human errors in scanning/counting)	Low (fully automated with fixed logic)
Future Scalability	Limited	High (can be IoT/cloud-enabled later)

Hence to overcome these drawbacks we have designed “The Smart Shopping Trolley with Automated Billing”that integrates RFID (Radio Frequency Identification), sensors, and a screen display to enable real-time billing, reduce human error, and offer a faster, smarter, and more convenient shopping experience.

3.2 Proposed Block Diagram

In this chapter the block diagram of the project and design aspect of independent modules are considered. Block diagram is shown in fig: 3.

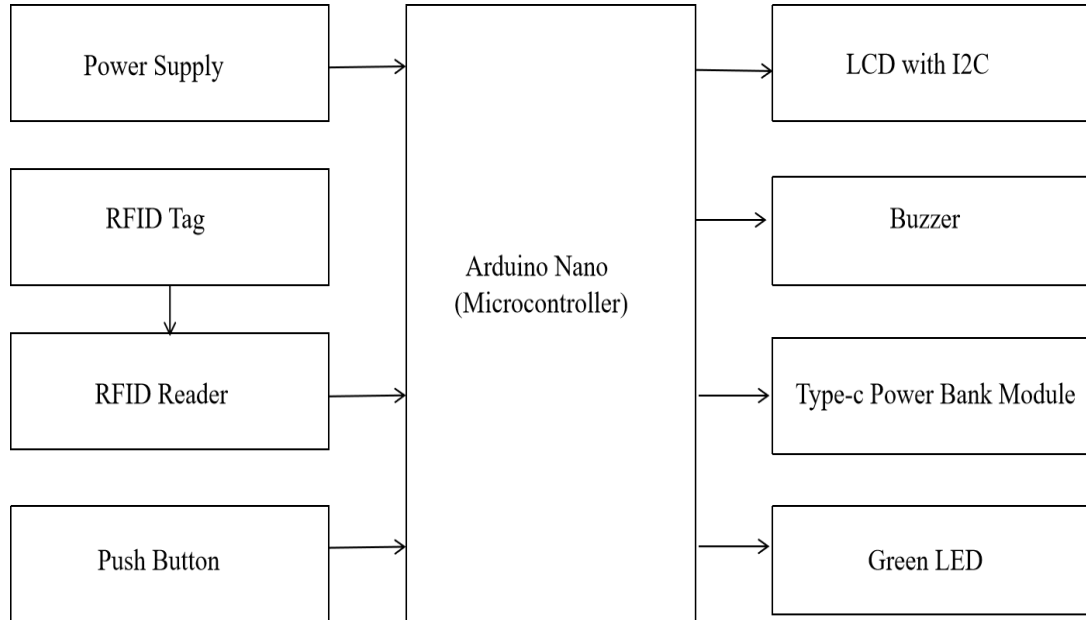


Fig. 3.1 Proposed Block Diagram

The block diagram illustrates the architecture of the Smart Shopping Trolley system, where the Arduino Nano acts as the central control unit. All essential components are interfaced with this microcontroller to perform specific functions.

The system is powered by a rechargeable 18650 Li-ion battery, which supplies energy to the Arduino and other peripherals through a Type-C charging module. An RFID reader is connected to the Arduino to detect RFID tags embedded on shopping items. When an RFID tag is brought near the reader, its data is sent to the Arduino for identification and billing. A push button is integrated to allow item removal functionality; when pressed during a rescan of the same item, the system identifies it as a removal request. The Arduino processes these inputs and updates the current bill on a 16x2 LCD display using an I2C interface for efficient communication. Additionally, the system includes a buzzer and a green LED to provide audio-visual

feedback during scanning or removal of items. Each component plays a critical role in making the trolley interactive, user-friendly, and capable of performing real-time billing operations without external intervention. The layout ensures smooth communication among components, low power consumption, and portability, making it suitable for practical implementation in supermarkets.

3.3 Flow Chart with Working Principle

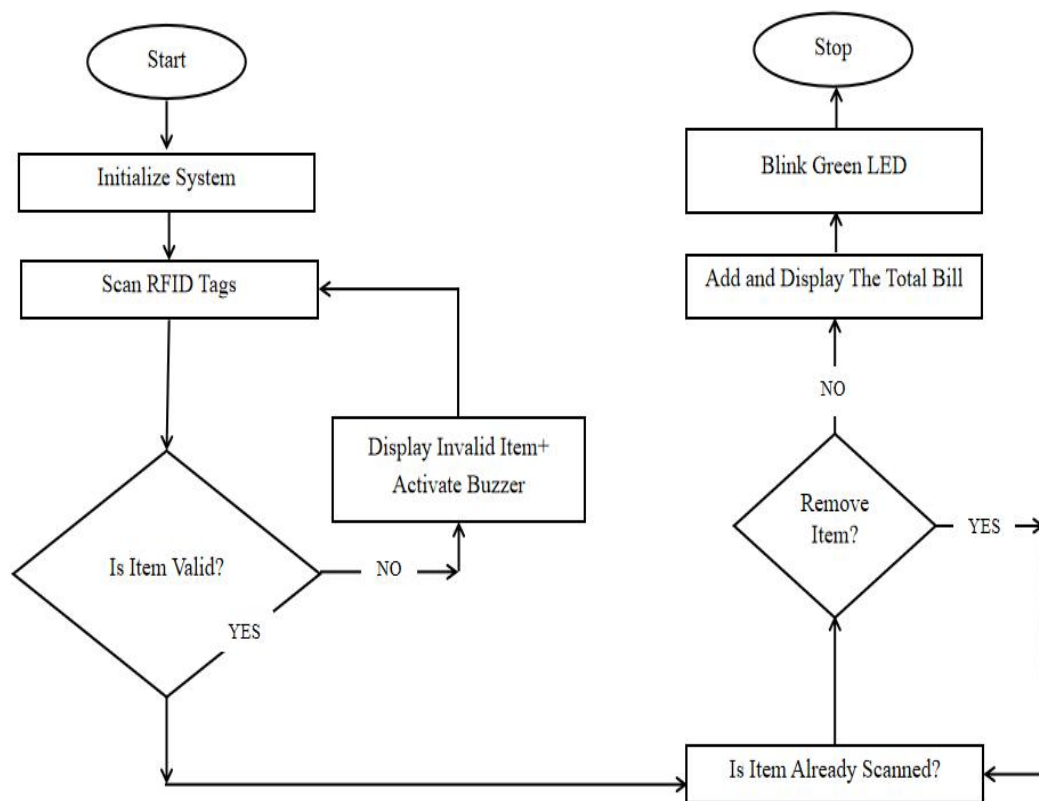


Fig. 3.2 Flow Chart

The flowchart represents the complete operational logic of the Smart Shopping Trolley system, starting from initialization to final billing. When the system is powered on, the microcontroller (Arduino Nano) initializes all components such as the LCD display, RFID reader, push button, and buzzer. Once initialized, the system enters an active state where it constantly waits for RFID tags to be scanned by user.

When a product is brought near the RFID reader, the tag's unique ID is detected and sent to the microcontroller for validation. If the scanned item is not listed in the database or is unrecognized, the system identifies it as an invalid item. In such cases, it immediately displays an error message on the LCD and activates the buzzer to alert the user. This prevents any unregistered or faulty tags from being added to the bill.

If the RFID tag is valid, the system checks whether the item has already been scanned earlier. If not, it considers it a new item and adds the product to the cart. The item name and price are shown on the LCD, and the total bill is updated in real-time. If the user scans an item that is already in the list, the system further checks if the push button is being pressed. If yes, it understands that the user wants to remove that item from the cart. In this case, the item is deleted from the list, the total is adjusted accordingly, and a feedback buzzer sound is played for confirmation.

The system continues to accept scans until the user is done with shopping. Once all desired items are scanned and the user stops interacting, the system adds up the final bill, displays it on the LCD screen, activates the buzzer one last time, and blinks the green LED to indicate that the shopping session is complete. This entire process allows customers to manage item scanning, removal, and billing independently, thus making shopping faster, more convenient, and free from billing queues.

CHAPTER 4

HARDWARE REQUIREMENTS

4.1 Components

Below is a detailed explanation of the minimal yet crucial hardware setup used in the development and testing of this project.

4.1.1 Arduino Nano

The Arduino Nano serves as the central control unit of the Smart Shopping Trolley system. It is a compact, breadboard-friendly microcontroller board based on the ATmega328P chip, offering sufficient I/O pins, memory, and performance for small-scale embedded projects like this one.

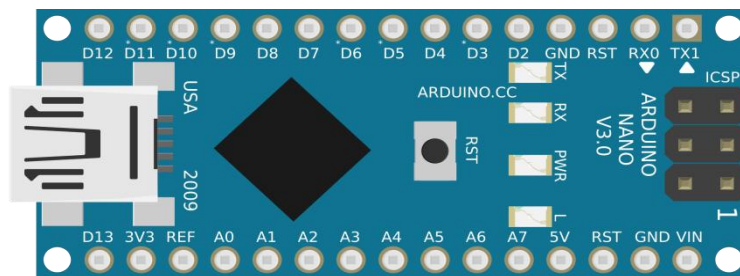


Fig. 4.1.1 Arduino Nano

In this project, the Arduino Nano is responsible for managing all the core operations. It receives data from the RFID reader whenever an item is scanned, processes that data to determine if the item is valid or already scanned, and then updates the LCD display accordingly with the item details and total cost. It also reads inputs from the push button, checks the status to see if an item needs to be removed, and handles corresponding changes in the billing logic. Furthermore, it controls output actions like activating the buzzer and blinking the green LED as feedback to the user. All logical decisions, timing, and communication between input and output devices are handled by the Arduino Nano, making it the backbone of the system.

It operates at 5V and runs at a clock speed of 16 MHz, making it both power-efficient and reliable for real-time processing. Its small size and USB program make it ideal for compact projects like a smart trolley, where space and portability are important.

4.1.2 RFID Reader and RFID Tags

The RFID Reader (RC522) is a key component of the Smart Shopping Trolley system, responsible for scanning the RFID tags attached to each shopping item.



Fig. 4.1.2 RFID Reader

This module operates at 13.56 MHz and communicates with the Arduino Nano via the SPI protocol. When a customer brings an item close to the reader, it detects the unique ID stored on the RFID tag and sends it to the Arduino for processing. The Arduino then compares the scanned tag ID with the predefined database to identify the item, add it to the bill, or remove it if needed.

The RFID system ensures contactless and quick item detection, allowing for a smooth user experience without the need for line-of-sight scanning like in barcode systems. Each product in the trolley is equipped with a unique RFID tag, which acts like a digital identifier. These tags are passive, meaning they don't require a power source and get activated when brought near the reader. This combination of the RC522 reader and RFID tags enables real-time item management, precise billing, and efficient automation of the shopping process.

4.1.3 16x2 LCD with I2C Interface

The 16x2 LCD with I2C interface plays a crucial role in the Smart Shopping Trolley system by providing a user-friendly display of scanned items and the running total bill. This module is a standard 16-character, 2-line liquid crystal display that is enhanced with an I2C (Inter-Integrated Circuit) adapter, which reduces the number of pins required for communication from 12 to just 4.

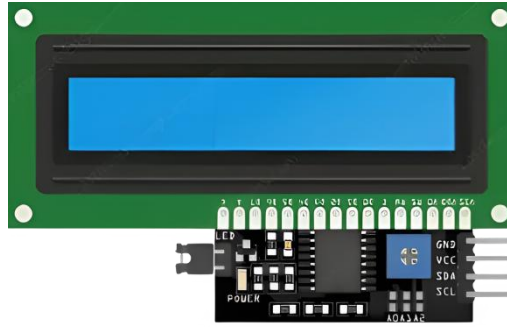


Fig. 4.1.3 LCD with I2C Protocol

This makes it ideal for projects with limited I/O pins like the Arduino Nano. In this project, the LCD is used to display messages such as “Welcome to Supermarket”, item names, prices, and the total bill in real-time as the customer scans or items. It also shows alerts for invalid RFID tags or item removal confirmations. The I2C interface ensures that the display updates are fast and accurate while minimizing wiring complexity. This makes the LCD not only essential for visual interaction but also effective in maintaining a clean and compact circuit design.

4.1.4 Type-C Power Bank Module

The Type-C Power Bank Charging Module is used in this project to provide a reliable and rechargeable power source for the Smart Shopping Trolley system. It allows the 18650 Li-ion battery to be charged via a Type-C USB port, which is more modern and convenient compared to traditional micro-USB.

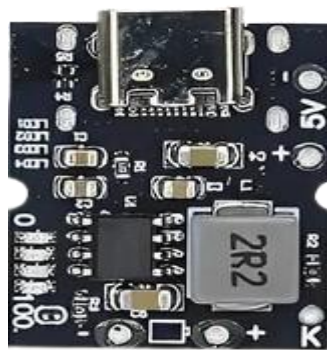


Fig. 4.1.4 Type-C Power Bank Module

This module includes a built-in battery protection circuit to prevent overcharging, over-discharging, and the need for a constant external power supply.

In this project, it supplies stable voltage to the Arduino Nano and other components,

making the entire system portable and eliminating the need for a constant external power supply.

4.1.5 18650 Li-ion Battery



Fig. 4.1.5 18650 Li-ion Battery

The 18650 Li-ion Battery is used as the primary power source for the Smart Shopping Trolley system. It is rechargeable, it powers the Arduino Nano and all connected components of the high-capacity battery commonly used in portable electronics. In this project, it powers the Arduino Nano and all connected components through the Type-C charging module. Its compact size, long life, and stable voltage output make it ideal for ensuring the system runs efficiently without frequent recharging.

4.1.6 Rocker Switch

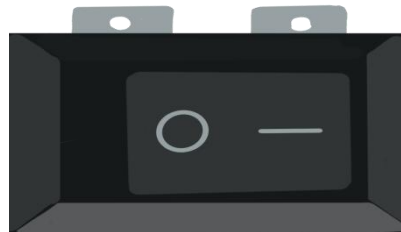


Fig 4.1.6 Rocker Switch

The Rocker Switch is used to turn the Smart Shopping Trolley system ON or OFF manually. It is simple, durable, and ensures safe control over the power flow from the battery to the components. It acts as the main power control, allowing the user to start or stop the circuit easily. It is simple, durable, and Smart Shopping Trolley system ensures safe control over the power flow from the battery to the components.

4.1.7 Push Button

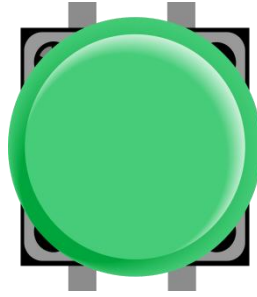


Fig 4.1.7 Push Button

The push button in this project is used to trigger the item removal function. When the user presses the button and simultaneously rescans an item, the system understands that the item should be removed from the cart. It acts as a simple digital input to the Arduino Nano and helps provide control to the user during shopping.

4.1.8 Buzzer



Fig 4.1.8 Buzzer

The buzzer in this project is used to provide audio feedback to the user during item scanning and removal. It is connected to the Arduino Nano and is activated whenever an invalid RFID tag is scanned or an item is successfully removed from the cart. This helps alert the user instantly, making the system more interactive and user-friendly.

CHAPTER 5

SOFTWARE REQUIREMENTS

5.1 Arduino IDE



The Arduino IDE is an open-source software platform developed specifically for writing and uploading programs to Arduino-compatible boards such as Arduino Nano, which is used in this project. It provides a simple and user-friendly interface that enables beginners as well as advanced users to develop embedded system applications using the Arduino programming language, which is based on C/C++.

The Arduino IDE consists of a code editor, a message console, a text output area, a toolbar with buttons for common functions (like verify, upload, new, open, save), and a serial monitor for real-time communication with the board. One of the most powerful features of the IDE is its ability to upload the code to the board via USB, allowing fast and seamless testing, debugging, and updates.

The structure of every Arduino sketch (program) consists of two main functions:

- `setup()` – which runs once and is used to initialize the system (e.g., setting pin modes, starting serial or LCD communication)
- `loop()` – which runs continuously, forming the heart of the program's logic and decision-making

In this Smart Shopping Trolley project, the Arduino IDE is used to write the program that reads RFID tag data through RC522 reader, updates the total cost on LCD screen, activates the buzzer when needed, removes items using a push button. The code is written in a clear structure with conditional if and else logic to perform tasks such as checking for repeated scans, validating tag IDs, and controlling hardware outputs.

Another major advantage of Arduino IDE is the built-in library manager, which allows you to include prewritten libraries like:

- SPI.h – for communication between Arduino and RFID reader
- MFRC522.h – for handling the RFID scanner module
- Wire.h – for I2C communication
- LiquidCrystal_I2C.h – for controlling the LCD display using I2C protocol

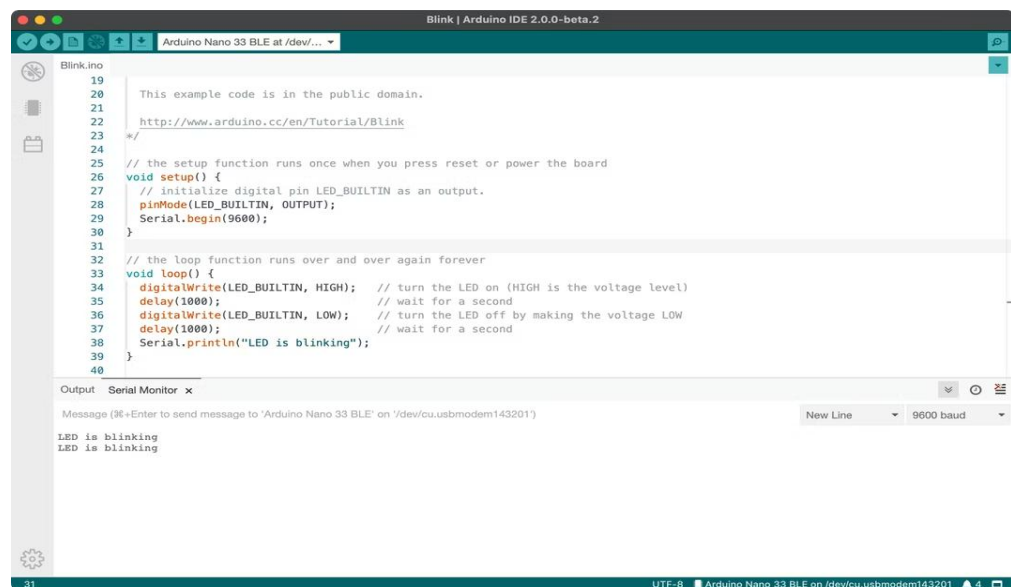


Fig 5.1 Arduino IDE Software Screen

The IDE also has a Serial Monitor and Serial Plotter, which are very useful during the development and testing phase. They allow you to monitor outputs, debug tag IDs, and understand how the program is functioning in real time.

The Arduino IDE is platform-independent, meaning it works on Windows, macOS, and Linux. It also supports third-party board installation, making it highly flexible and scalable. In this project, the board selected in the IDE is “Arduino Nano”, and the programmer can directly upload the compiled .hex code to the microcontroller via the USB connection. Arduino IDE serves as the bridge between the programmer and the physical hardware. Without it, writing and uploading code to the Arduino Nano would not be possible. It provides the tools to design logic, control hardware operations, debug system behavior, and bring the Smart Shopping Trolley project to life.

CHAPTER 6

RESULTS

The Smart Shopping Trolley was tested after successful integration of both hardware and software components. After uploading the program to the Arduino Nano using the Arduino IDE and connecting all components, the system was powered using a 18650 Li-ion battery. The project responded as expected during real-time testing.

When the trolley was switched on, the LCD display showed the welcome message “Welcome to Supermarket – Scan Item.” Upon scanning an RFID-tagged item, the system identified the tag correctly, displayed the item name and price on the LCD, and added it to the total bill. The bill automatically updated with each new scan. If the same item was scanned again while holding the push button, the system treated it as a removal request and subtracted the item’s price from the total. The buzzer provided an audio alert after each scan and removal action, while the green LED blinked at the end to confirm the billing was complete.

The system performed successfully in all test scenarios, including multiple item scanning, item removal, and total calculation. The real-time display on the LCD worked without delay, and the item management was smooth and accurate. Photographs of the output screens were captured during testing and are presented in this section for verification.

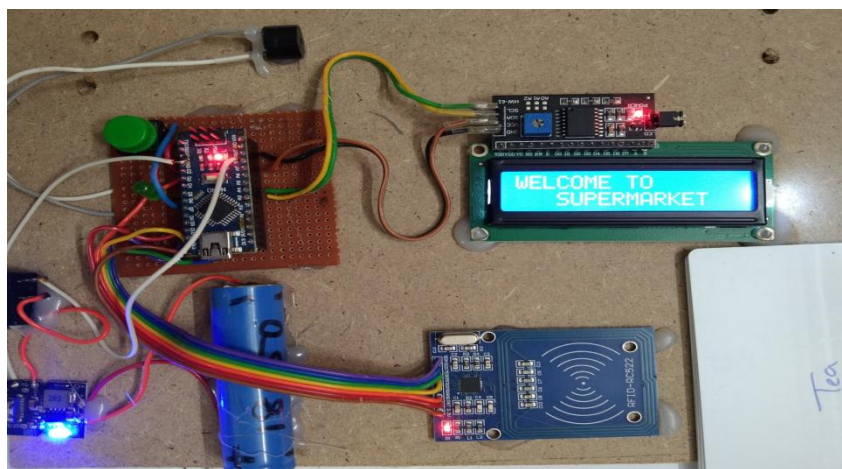


Fig 6.1: Welcome Message on System Initialization

Once the system is powered on, the LCD displays a welcome message “WELCOME TO SUPERMARKET - SCAN ITEM” indicating that the system has been initialized and is ready for scanning items. This confirms that the microcontroller, LCD, and power module are working properly.

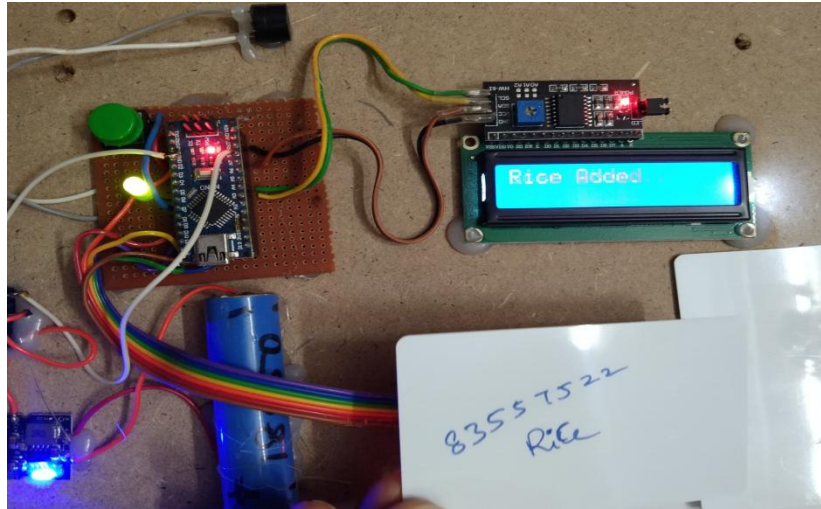


Fig 6.2: Adding Item - Rice Scanned

The item “Rice” is scanned. The microcontroller reads the RFID tag, verifies it, and adds it to the bill. The LCD confirms “Rice Added.” The green LED glows briefly to indicate successful addition of the item.



Fig 6.3: Adding another Item - Butter Scanned

When the RFID tag labeled "Butter" is scanned, the system reads the unique ID and maps it to the item stored in the program. The LCD displays “Butter Added” along with the total price updated to Rs.171.00.

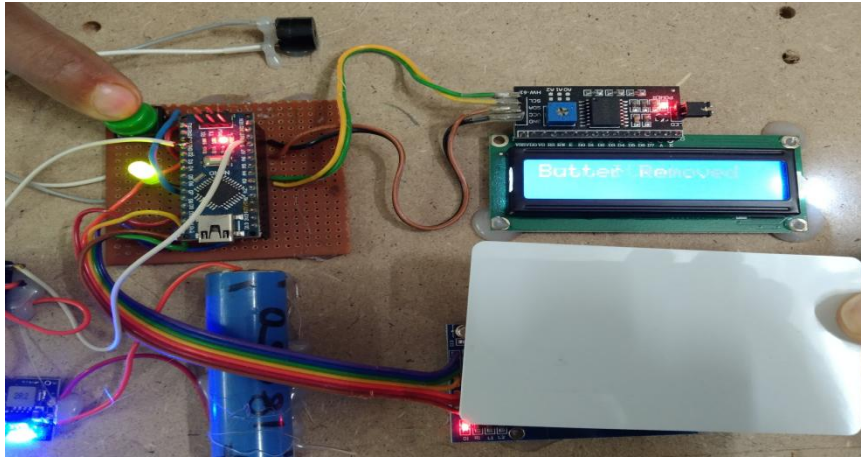


Fig 6.4: Removing Item

Since "Butter" was scanned twice by mistake, the pushbutton is pressed while scanning the same RFID tag again. This triggers the removal function, with the LED glowing.

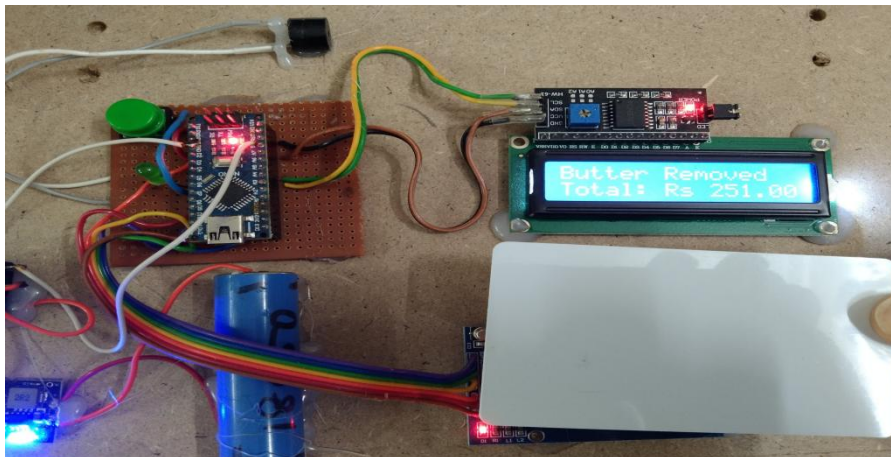


Fig 6.5: Removing Duplicate Item - Butter Removed

The system subtracts the item price from the total. The LCD displays “Butter Removed” and the corrected total of Rs.251.00.

CHAPTER 7

CONCLUSION

7.1 Conclusion

The Smart Shopping Trolley with Automated Billing project represents a significant step forward in the evolution of retail automation and customer convenience. The aim of this project was to design and develop an intelligent shopping trolley that can automate the billing process using RFID technology while minimizing the need for manual intervention during checkout. After a thorough implementation of both hardware and software components, the system successfully demonstrated the ability to perform real-time item detection, total billing, item removal, and final display of charges in a seamless and user-friendly manner.

By integrating an RFID module, Arduino Nano microcontroller, LCD with I2C, buzzer, push button and battery-powered power system, the prototype was able to function as an independent smart billing unit. As soon as the RFID-tagged items are scanned, they are added to the cart and displayed on the LCD screen, with the total cost updated in real-time. The use of a push button allows customers to easily remove items if they change their minds during shopping. All operations are designed to be simple, efficient, and responsive, thereby improving the overall shopping experience. This system effectively eliminates one of the biggest bottlenecks in supermarkets — the long queues at billing counters. Since the billing takes place as the customer shops, it ensures that by the time the shopping is done, the final amount is ready for payment. This self-checkout model not only saves time but also reduces the manpower needed at counters, which is beneficial for store owners as well.

The development of this project also allowed for an in-depth exploration of microcontroller programming, sensor integration, and embedded system design. Challenges like power management, real-time updating of the LCD display, and scanning validation logic were successfully addressed during implementation. The use of an 18650 Li-ion battery with a Type-C charging module ensured portability and ease of use without the need for external power sources.

In conclusion, this project not only fulfilled its initial objectives but also opened up new possibilities in the field of smart retail systems. The implementation proved to be technically sound, cost-effective, and highly relevant to the current demand for automation in the retail sector.

With further development and refinement, the Smart Shopping Trolley can become a real-time solution for modern shopping environments, improving efficiency for businesses and enhancing the overall satisfaction of customers.

7.2 Future Scope

The Smart Shopping Trolley system, though currently implemented as a basic prototype, has strong potential for future upgrades and real-world deployment. In the future, this project can be enhanced by integrating wireless communication modules such as Wi-Fi or Bluetooth to connect the trolley with centralized billing systems or cloud-based inventory databases. This would allow real-time data synchronization with store servers, enabling features like digital receipts, live stock tracking, and customer purchase history.

The system can be extended to support mobile payment gateways or QR-code-based checkout, making the entire shopping experience contactless. Features like voice assistance, touchscreen interfaces, or automatic item detection using weight sensors or cameras can be introduced for improved accessibility and security. Integration with store mobile apps can also provide personalized offers and navigation within the store. With these improvements, the Smart Shopping Trolley can evolve into a fully functional component of the next-generation smart retail infrastructure, suitable for deployment in supermarkets, hypermarkets, and even unmanned stores.

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APPENDIX

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define SS_PIN 10
#define RST_PIN 9
#define BUTTON_PIN 6
#define RED_LED 8
#define GREEN_LED 5
#define BUZZER 4
MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2); // Adjust I2C address if needed
double total = 0;
int count_prod = 0;
int p1 = 0, p2 = 0, p3 = 0, p4 = 0, p5 = 0;
// Replace with your actual tag UUIDs (use Serial.print to find them)
byte butterUUID[4] = {0x03, 0xD2, 0x18, 0x22};
byte milkUUID[4] = {0xC3, 0xCB, 0x2E, 0x22};
byte teaUUID[4] = {0x73, 0xD1, 0x1C, 0x22};
byte riceUUID[4] = {0x83, 0x55, 0x75, 0x22};
byte soapUUID[4] = {0x13, 0x67, 0x2B, 0x22};
void setup()
{
  Wire.begin();
  lcd.begin();
  lcd.backlight();
  SPI.begin();
  mfrc522.PCD_Init();
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  pinMode(RED_LED, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);
```

```

pinMode(BUZZER, OUTPUT);
lcd.setCursor(0, 0);
lcd.print("AUTOMATIC BILL");
delay(2000);
lcd.clear();
lcd.print("SHOPPING CART");
delay(2000);
lcd.clear();
lcd.print("WELCOME TO");
lcd.setCursor(3, 1);
lcd.print("SUPERMARKET");
delay(2000);
lcd.clear();
lcd.print("Scan Item...");
}
void loop()
{
if (!mfrc522.PICC_IsNewCardPresent() || !mfrc522.PICC_ReadCardSerial())
return;
byte* scannedUID = mfrc522.uid.uidByte;
bool removeMode = digitalRead(BUTTON_PIN) == LOW;
if (!removeMode) p1++; else if (p1 > 0) p1--;
}
if (compareUID(scannedUID, butterUID))
{
handleItem("Butter", 47.0, p1, removeMode);
else if (compareUID(scannedUID, milkUID))
{
handleItem("Milk", 33.0, p2, removeMode);
if (!removeMode) p2++; else if (p2 > 0) p2--;
}
else if (compareUID(scannedUID, teaUID))
{
handleItem("Tea", 29.0, p3, removeMode);

```



```

    if (!removeMode) p3++; else if (p3 > 0) p3--;
    }
    else if (compareUID(scannedUID, riceUID))
    {
        handleItem("Rice", 50.0, p4, removeMode);
        if (!removeMode) p4++; else if (p4 > 0) p4--;
    }
    else if (compareUID(scannedUID, soapUID))
    {
        handleItem("Soap", 45.0, p5, removeMode);
        if (!removeMode) p5++; else if (p5 > 0) p5--;
    }
    else
    {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Unknown Item!");
        buzz(300);
        delay(2000);
    }
    mfrc522.PICC_HaltA();
    mfrc522.PCD_StopCrypto1();
}

void handleItem(const char* name, float price, int quantity, bool removeMode)
{
    lcd.clear();
    if (removeMode)
    {
        if (quantity > 0)
        {
            lcd.print(name);
            lcd.print(" Removed");
            total -= price;
            count_prod--;
        }
    }
}

```

```

buzz(300);
}
else
{
    lcd.print("Not in cart!");
    buzz(300);
}
}
else
{
    lcd.print(name);
    lcd.print(" Added");
    total += price;
    count_prod++;
    buzz(300);
}
lcd.setCursor(0, 1);
lcd.print("Total: Rs ");
lcd.print(total);
delay(4000);
lcd.clear();
lcd.print("Scan Item...");
}

bool compareUID(byte* a, byte* b)
{
    for (byte i = 0; i < 4; i++)
    {
        if (a[i] != b[i]) return false;
    }
    return true;
}

void buzz(int duration)
{

```

```
digitalWrite(RED_LED, HIGH);  
digitalWrite(GREEN_LED, HIGH);  
digitalWrite(BUZZER, HIGH);  
delay(duration);  
digitalWrite(RED_LED, LOW);  
digitalWrite(GREEN_LED, LOW);  
digitalWrite(BUZZER, LOW);  
}
```